AI Based Intelligent Truck Monitoring and Risk Assessment System.

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***Abstract*—Manual observation and identification, speed checking and security verification with disconnected tools are challenging to conduct surveillance on trucks. On this purpose an intelligent monitoring system and risk assessment of trucks is proposed in this paper, that integrates all the above tasks in a single computer vision pipeline for a particular truck. The system identifies and tracks trucks in real-time with live CCTV feed, calculates the truck speed, identifies truck license plate via Automatic License Plate Recognition (ALPR), and categorizes the truck type, colour and the visual characteristics by using a multi-task vision-transformer model. Real-time detection and alert of overspeeding, blocklisted and suspicious vehicles is achieved through a variety of means and recognised vehicle plates are checked against a blocklist. All data is collected in MySQL, is served by a Flask REST API and is presented in a React dashboard for live monitoring, search and analytics. The proposed system is not just a single purpose detection or ALPR pipeline reported in the literature but it integrates detection, tracking, recognition and risk based alert in a single tracked identity.**

***Index Terms*—object detection, automatic license plate recognition, risk assessment, and intelligent transportation systems.**

I. INTRODUCTION

Road freight monitoring is currently mostly undertaken by an operator, who monitors a camera feed, manually records registration numbers and verifies another road user authorisation system (where available), whilst speed monitoring is completed by another system (where available). These functions are independent of each other and when a vehicle is overspeeding and flagged, it might not get detected if there is heavy traffic, and/or operator fatigue. Although each of these has evolved independently, none of these systems is integrated with the detection, tracking and LPR process with a single identity carried across all three layers to a single risk decision. In this paper such a system is proposed, tailored to the needs of trucks. It has three parts to it: the tracking identity as to which it lives, the risk-fusion stage where an overspeed signal, blocklist signal and attribute signal are fused together to make a decision, and the surrounding database, API and dashboard where the detection output is fed, so that the operator can monitor and act on it.

A. Problem Statement

Manual observations and isolated, unconnected, and "silent" systems to monitor vehicles form a significant part of the traditional truck monitoring system, with the cost of this being that it slows down the ability to detect overspeed vehicles, suspicious vehicles, or when a vehicle's observed characteristics are not matching the information entered into the system. The current literature is largely devoted to individual tasks, but not specifically to a comprehensive risk assessment process specific to the truck. This project seeks to close the gap and will involve the development of a computer vision system capable of recognising and tracking trucks in real time, recognising plates, verifying attributes, analysing speed and security status and issuing alerts.

B. Objectives

The project develops the following pipeline of visual AI models: a lightweight truck detector model that can detect trucks in real time on real streams; a multi-object tracker model that can continuously identify each truck in each frame and a speed-estimation model that receives these information and can have an overspeed threshold that can be programmed. ALPR module uses PaddleOCR to process number plates in various lighting conditions and angles to identify the number plates, and multi-task classifier classifies the truck type, colour and other visual properties of the truck in one shot. Recognised plates are matched against a list of vehicles considered flagged. The overspeed result, blocklist match and any difference from the observed data and registered data are input to a risk-fusion engine to make one decision. If this decision requires action the system informs authorised personnel as soon as they are needed via sound, dashboard alerts, and via email or SMS communication. All of these records are kept in a MySQL database, then served to a Flask REST API, which is connected to a React dashboard where they can be monitored, searched and analysed.

C. Research Gap

No existing system in existence has been surveyed which combines all the following components into a single pipeline: Detection, Persistent Tracking, ALPR, Attribute Classification, Speed Estimation, Blocklist Verification and Risk Based Alerting.

The focus of detection/tracking papers is not on the classification of trucks, but rather on general vehicle data sets; not on searching a truck registry.

ALPR papers are not “persistently identifiable”, so that a plate can't be linked to a vehicles actions over a period of time or their speed.

All the systems reviewed lack a database based blocklist check, multi-channel alert and a fused risk decision.

II. LITERATURE SURVEY

Six recent publications from IEEE Journal were analyzed, summarised and described along with their advantages, disadvantages and their comparison with the proposed work are given below.

To improve the YOLOv8 detector and BoT-SORT tracking on UA-DETRAC, this paper is based on the base paper (Liu & Shen, 2025) [1] to propose an enhanced version of YOLOv8 and BoT-SORT tracking. The lightweight YOLOv8n detector boosts multi-vehicle tracking accuracy, but is only capable of vehicle detection and tracking and has no ALPR, classification or risk logic. This is considered as a baseline of the perception part of the proposed system and is further extended to include ALPR, classification, block list verification, risk fusion and alerting.

An IEEE ICISC (2025) paper on integrated LPR using YOLO and CNN [2] combines the plate localisation model YOLOv8 with a CNN and EasyOCR model to read the plate characters, allowing to achieve the complete plate detection and reading in one pipeline. However, is not sustained on tracking or risk assessment and hence there is no link to vehicle behaviour and a plate reading. Proposed system on the contrary provides each plate reading a persistent tracked identity and a risk decision.In the IEEE Sensors Journal 2024, vehicle classification and LPR using deformable YOLOv8 [3] improves YOLOv8 for classification and plate reading in challenging lighting and weather conditions, resulting in strong detection and classification under challenging visual conditions. There is no tracking, speed estimation, checking of blocklists or signal fusion. This is the missing persistent tracking, speed estimation and risk fusion that the proposed system provides.

Comparative accuracy data is provided on LPR using YOLOv8 vs. EasyOCR and PaddleOCR vs. Tesseract for plate character recognition to aid in selecting an OCR-engine [4] is published in 2024 IEEE. This study focuses on benchmarking and does not consider monitoring workflow or vehicle-identity tracking. It helped determine the selection of PaddleOCR for the proposed ALPR module.

To achieve high accuracy for recognizing the text on the plates, an IEEE CITSM (2023) paper that integrates a YOLOv8 detector with EasyOCR achieves around 95% plate-recognition accuracy on the curated test dataset [5]. It provides no tracking of trucks and no risk logic, just ALPR. The proposed system integrates such ALPR accuracy into a complete risk assessment system.

Around 94% accuracy is achieved for ALPR feasibility, as shown from the IEEE ICOIACT (2023) study of Indonesian local plates data set comprising 869 images for ALPR system based on YOLOv8 with EasyOCR [6]. Small size of dataset and lack of end-to-end monitoring or classification. The proposed system generalises ALPR into a larger, multi-module truck-monitoring pipeline.

A brief review of the various surveys and overall comparison.

The surveyed papers are broken into two categories: Detection/tracking papers which were evaluated based on general criteria like UA-DETRAC [1] (plate accuracy 93-95%), and ALPR papers which combined an ALPR detector with an OCR engine (plate accuracy 93-95%) [2],[4]-[6]. There is no connection to a plate reading or a vehicle's speed and there is no registry check. None of the surveyed papers fuse more than two of persistent tracking, ALPR, attribute classification, blocklist/risk fusion, and real-time alerting; the proposed system is behind one tracked identity.

III. SYSTEM ARCHITECTURE

A. Architecture Diagram

The system is structured into a 13-module pipeline (Fig. 1) into four layers, input/perception, parallel analysis, decision/alerting, and data/presentation.

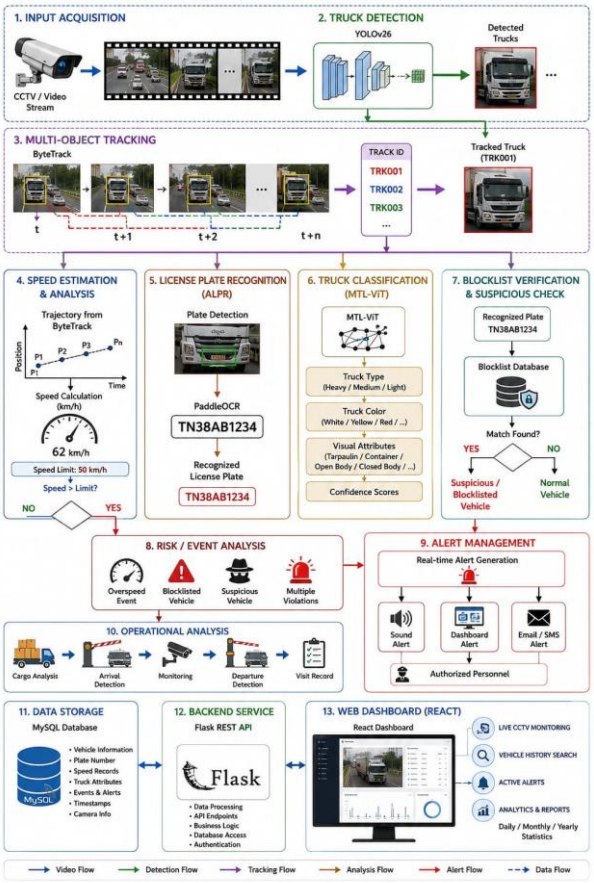


Fig. 1. Proposed architecture of the intelligent AI-based truck monitoring system.

B. Proposed Methodology

FPS from a CCTV/IP camera are fed into a YOLO-based truck detector, and the detections are passed to the ByteTrack multi-object tracker, where every vehicle is given a stable Track ID that is maintained across the frames. Based on the tracked trajectory, the speed is estimated by calculating the

displacement of the centroid and compared to a given threshold, while at the same time, the PaddleOCR system reads the plate from the tracked crop and an MTL-ViT module classifies the truck type, colour and attributes using a shared multi-task head. The recognized plate is compared to a blocklist and any overspeed, blocklist or suspicious-attribute signal is fused with the other signals to produce a multi-channel alerting signal, meaning that if a registered plate is detected to have more than one signal, the vehicle is escalated from the risk signal to the multi-channel signal.

C. System Design

Perception modules (detection, tracking) are applied on each frame in the video and are limited to lightweight functions; analysis modules (speed, ALPR, classification, blocklist check) are applied to each tracked vehicle and run concurrently as they do not depend on the others results. The data/presentation layer (MySQL, a Flask REST API, and a React dashboard) maintain thresholds and alert policy, and stores and presents the content generated by the previous layers (decision modules: risk fusion, alerting, operational analysis).

D. Dataset and Experimental Setup

A dataset of images and video frames of various truck types, colours, angles and traffic scenes was divided into training and held-out testing sets, and feature annotations were provided for the truck bounding boxes, plate text and attributes. The system was developed in Python language (YOLO, ByteTrack, PaddleOCR, MTL-ViT, OpenCV), and deployed with a Flask/MySQL backend, and deployed using a React dashboard, and evaluated with recorded CCTV footage. The following measures were used to evaluate performance: mAP@0.5, ID-switch count, plate-recognition accuracy, classification accuracy, speed-estimation MAE and FPS.

IV. RESULTS AND DISCUSSION

This section tests the proposed system on unseen test data from the truck dataset, and on video recorded from CCTV footage. The first part of the report of the module level performance is shown in Table I: mAP@0.5 for detection, ID-switch count for tracking, plate accuracy for ALPR, classification accuracy for MTL-ViT, mean absolute error for speed estimation and frames per second for the full pipeline. The following illustrations (Figs. 2–7) present one tracked truck with its plate, speed and attributes, followed by the corridor-level analytics aggregated from many such records.

A. Quantitative Results

TABLE I. PERFORMANCE EVALUATION OF THE PROPOSED SYSTEM WAS DONE

Module	metric	Result
Truck Detection (YOLO)	mAP@0.5	91.8%
Multi-Object Tracking (ByteTrack)	ID Switches	5
ALPR (PaddleOCR)	Plate Accuracy	94.2%
Truck Classification (MTL-ViT)	Classification accuracy	89.7%
Speed Estimation	MAE	2.8 km/h
Complete System	Processing Speed	18 FPS

The detection rate of mAP@0.5 is 91.8% and the number of ID switches is 5 for ByteTrack. The plate accuracy rate of ALPR was 94.2% while most of the error was due to low resolution, motion blur and lighting; the classification accuracy rate of MTL-ViT was 89.7%. The overall performance of the speed estimation is a 2.8 km/h MAE, while the full pipeline is 18 FPS, appropriate for live monitoring.

B. Visual Results

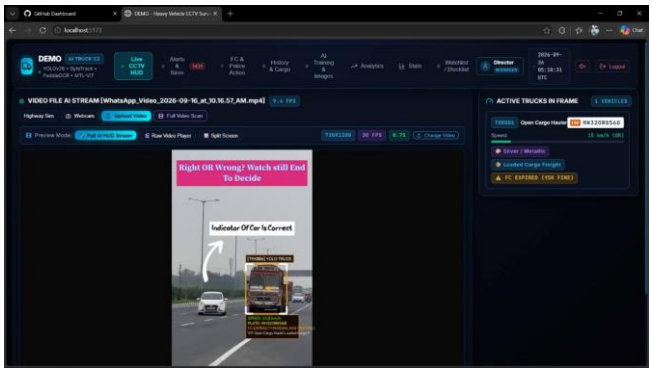


Fig. 2. Live CCTV HUD with tracked truck (TRK006) with speed and attribute overlay and plate.

The HUD detects, tracks, ALPR and classifies the plate, calculates an estimated speed (within the speed limits), provides colour and cargo properties, and issues an expired-fitness-certificate flag and fine — all in a single tracked identity, all in real-time.

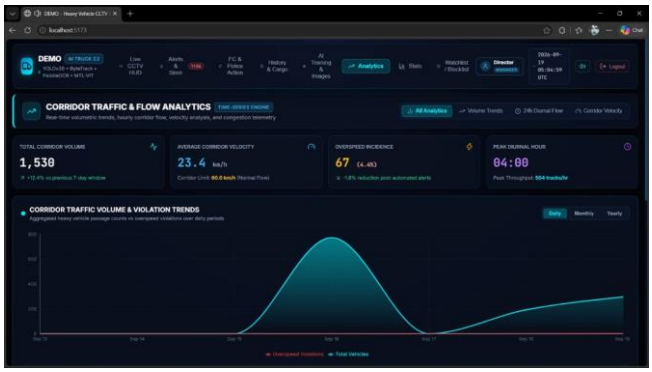


Fig. 3. Seven-day rolling violation & corridor trend analysis.

Records per vehicle are aggregated over time to get corridor-level KPIs, such as total volume, average velocity, overspeed incidence, and peak diurnal hour, with a daily trend line (volume/violation relationship) tracked by the operational-analysis module.

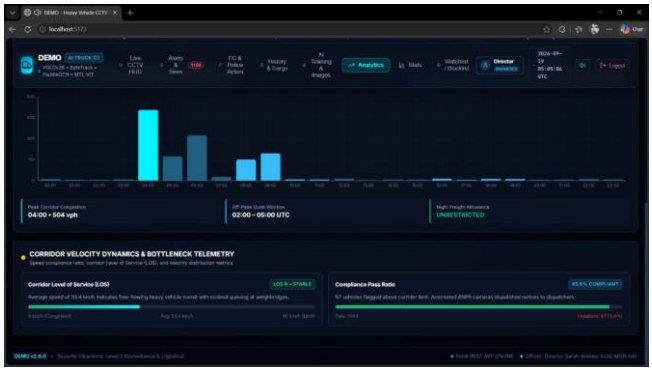


Fig. 4. Corridor velocity dynamics: Hourly throughput histogram and Level-of-Service and compliance-pass-ratio panels.

The level of congestion and off-peak times are displayed in an hourly histogram, the number of vehicles that were within the speed limit and the number of vehicles that were exceeding the speed limit are summarised in an ‘over speed' gauge and ‘compliance pass ratio' panel, respectively, reflecting the speed-estimation module's ‘overspeed' decision.

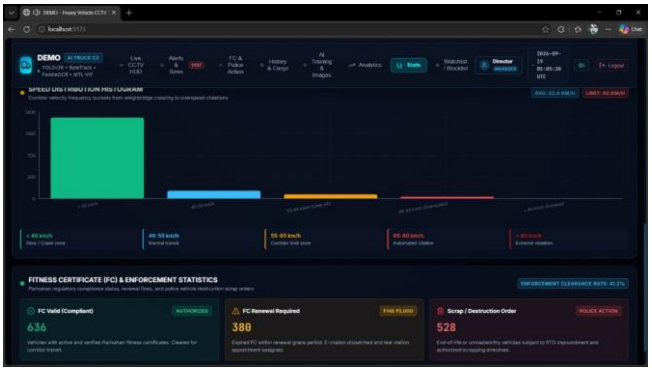


Fig. 5. Enforcement statistics for speed-distribution histogram or fitness-certificate (FC).

A speed signal and a registry-status signal are combined into a risk-fusion stage to bucket vehicles by speed band, as well as FC compliance counts (valid, renewal-required with fine, and scrap/destruction orders) for the same tracked truck.

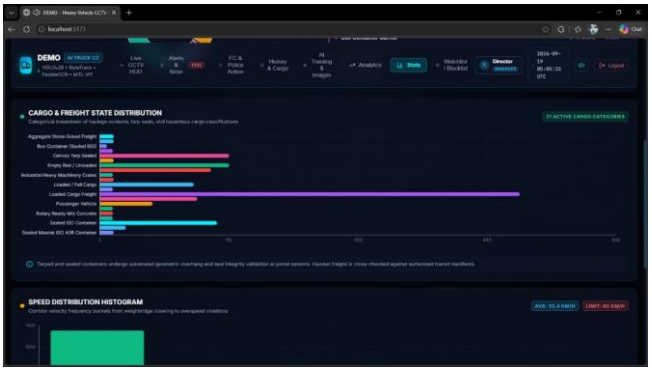


Fig. 6. Distribution of cargo and freight-state per classified cargo category.

The cargo-state predictions of the MTL-ViT attribute are aggregated here, in support of the automated overhang and seal-integrity checks, mentioned in the operational-analysis module.



Fig. 7. Fleet composition – vehicle-type classification in MTL and categorized census table.

The classification census verifies that the multi-task head is able to classify 15 different vehicle-type classes on the fleet of vehicles that are tracked, which is reflected in the classification-accuracy figure reported in Table I.

C. Discussion

Figs. 2–7 show that detection, tracking, ALPR, classification and registry-status signals are fused under a single tracked identity and reported on the operational dashboard as one signal, instead of requiring manual cross-checking between different tools. The accuracy of the results in Table I would be better substantiated on a larger sample of trucks and a more representative sample of trucks geographically.

V. CHALLENGES FACED

Few labels are provided for the classification module specific to the truck. The lighting, weather conditions, and camera angle influence the ability to accurately detect and read for a truck. The tracked vehicle identity can be lost due to occlusion and ID switching in dense traffic. Multiple concurrent modules to optimize for real time and accuracy is difficult. The modules have been developed in a modular fashion and must be connected to one pipeline. Another difficulty in the process of testing a real-world blocklist verification mechanism is the lack of an official blocklist for testing purposes.

VI. CONCLUSION AND FUTURE WORK

It suggested an intelligent truck monitoring and risk evaluation system based on AI, which integrated the detection, continuous monitoring, speed estimation, ALPR, multi-task classification and verification with the blocklist, and real-time alerting by using the same vehicle ID, but the current literature researched did not consider the combination of these abilities to be presented. Remaining works include, but are not limited to, the risk-fusion and alerting layers, evaluation on a large dataset of purpose collected trucks and extending to multiple concurrent camera streams with cross-camera re-identification.

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